

EMILY JIN

emily.jin@cs.ox.ac.uk • <https://ejin700.github.io>

EDUCATION

University of Oxford

Ph.D. in Computer Science, supervised by Michael Bronstein and İsmail Ceylan

Oxford, UK

Expected 10/2025

Columbia University, School of Engineering and Applied Science

M.S. in Computer Science

New York, NY

5/2021

B.S. in Computer Science, Minor in Chemical Engineering

5/2019

RESEARCH AND PROFESSIONAL EXPERIENCE

University of Oxford (collab. with AstraZeneca)

Doctoral Researcher, Dept. of Computer Science

Oxford, UK | Cambridge, UK

6/2022 – present

- Design large generative models to predict stable organic crystal structures for materials discovery
- Leverage biomedical knowledge graphs and path-based models to predict likely disease targets
- Constructed informative positional encoding kernel for Graph Transformers using motif structures
- Developed precise and expressive graph neural network architectures with graph homomorphisms
- Redefined path constraints to improve reinforcement learning models for drug repurposing by 16.4%

Microsoft Research

Research Intern, Health Futures

Redmond, WA

6/2024 – 9/2024

- Generated wholistic patient embeddings for rare disease diagnosis using biomedical knowledge graphs
- Clustered patient cohorts using geometric deep learning for causal gene discovery in rare genomics

University of Oxford (collab. with Merck KGaA)

Doctoral Researcher, Dept. of Engineering Science, Alison Noble Group

Oxford, UK | Geneva, CH

3/2022 – 6/2022

- Explored the creation of automated image diagnostic solutions for female genital schistosomiasis (FGS)
- Assessed the use of ResNet and attention-based CNN models for cervical cancer in relation to FGS

Facebook (now Meta)

Software Engineer, Solutions

New York, NY

9/2019 – 9/2021

- Released skippable video ads for Facebook and disclaimer ads in Instagram, unlocking \$6.5M revenue
- Designed tools to help advertisers simplify ads, reducing the number of pages above the limit by 38.2%
- Completed full-stack integration of ads recommendation tools, which received an 80% approval rating
- Migrated over 1,000 Facebook pages for pharma advertisers to remain in regulatory compliance

Columbia University, Irving Medical Center

Graduate Researcher, Dept. of Biomedical Informatics, Yufeng Shen Group

New York, NY

1/2019 – 5/2020

- Built a CNN model to predict the molecular effect of rare genetic mutations using protein regulation
- Increased model performance by 6.5% by including protein complex, structure, and expression features
- Used gradient class activation mapping to recover important sequence motifs and binding sites

PUBLICATIONS

- Linus Bao*, **Emily Jin***, Michael M. Bronstein, İsmail İlkan Ceylan, Matthias Lanzinger. *Homomorphism Counts as Structural Encodings for Graph Learning*. ICLR 2025. *Equal contribution.
- **Emily Jin**, Michael M. Bronstein, İsmail İlkan Ceylan, Matthias Lanzinger. *Homomorphism Counts for Graph Neural Networks: All About That Basis*. ICML 2024.
- **Emily Jin**, J. Alison Noble, Mireille Gomes. *A Review of Computer-Aided Diagnostic Algorithms for Cervical Neoplasia and an Assessment of Their Applicability to Female Genital Schistosomiasis*. Mayo Clinic Proceedings: Digital Health (2023).
- Alexander Kitaygorodsky, **Emily Jin**, Yufeng Shen. *Predicting localized affinity of RNA binding proteins to transcripts with convolutional neural networks*. bioRxiv (2021). Reviewers' Choice at ASHG 2019.

TALKS AND PRESENTATIONS

- **International Chemical Congress of Pacific Basin Societies (Pacifichem)**, Honolulu, HI 12/2025
- **Women in Computer Science Keep It Real**, Oxford UK 5/2025
- **University of Cape Town**, Cape Town, ZA 2/2025
- **International Women's Day Panel**, Oxford, UK 2/2025
- **Graph Learning on Wednesdays (GLOW)**, Virtual 12/2024
- **AI for New Drug Modalities Workshop (AIDrugX) @ NeurIPS 2024**, Vancouver, CA 12/2024
- **Learning on Graphs Conference (LoG)**, Oxford, UK 11/2024
- **Algorithms, Dynamics, and Information Flow in Networks Seminar**, Virtual 3/2024

ACADEMIC ACHIEVEMENTS

- **UKRI EPSRC Studentship**, £200,000 value over 4 years 9/2021
- **NSF Graduate Research Fellowship Honorable Mention** 4/2021
- **MIT EECS Viterbi Fellowship (Declined)**, \$100,000 value over 1 year 2/2021
- **Columbia University MS Teaching Fellowship**, \$30,000 value over 2 years 1/2020
- **Columbia University Dean's List** 5/2019

TEACHING

Mediterranean Machine Learning Summer School (M2L)

Split, HR

Tutor, Graph Neural Networks (GNNs)

9/2025

- Create and lead tutorial exercises for GNNs and assist with LLM, Vision, RL, and Diffusion tutorials

University of Oxford

Oxford, UK

Tutor, Schmidt Science Fellows

1/2025 – 6/2025

- Privately tutored a Schmidt Science Fellow in Biology on graph neural networks and machine learning

Demonstrator, Dept. of Computer Science

10/2022 – 12/2024

- Designed and lead tutorials for Geometric Deep Learning and Graph Representation Learning courses
- Lectured for the department's Artificial Intelligence course to undergraduate and master's students
- Mentored student projects for Software Engineering and Data Science for Drug Discovery courses

Columbia University

New York, NY

Teaching Assistant, Dept. of Computer Science

9/2017 – 5/2021

- Held office hours for Machine Learning, Advanced Programming, Essential Data Structures courses
- Organized overnight student hackathon for over 400 students in Advanced Programming course

SERVICE

AI for Accelerated Materials Design Workshop (AI4MAT) @ NeurIPS 2025

San Diego, CA

Workshop Organizer

4/2025 – present

- Facilitate proposal writing, program creation, industry sponsorship, and organizational logistics

Oxford Women in Computer Science

Oxford, UK

Vice President

5/2024 – 5/2025

- Promoted women and gender minorities studying computer science and related STEM fields at Oxford
- Established society scholarships to support members who are presenting their research at conferences

Industry Officer

5/2023 – 5/2024

- Collaborated with industry sponsors to organize networking dinners, panels, and technical workshops

New College, Oxford

Oxford, UK

Middle Common Room (i.e. Graduate Student Body) President

5/2022 – 5/2023

- Represented the graduate student community at meetings with college administration and alumni
- Managed 20+ committee members to host academic and social events and improve graduate life